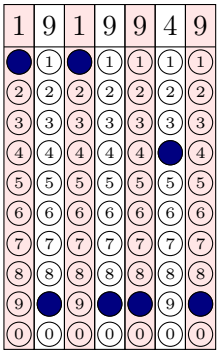
TEST VARIANTI RAQAMI: **1919949**

MATEMATIKA

Ushbu test varianti 30 ta test topshirig‘idan iborat. Uni bajarishga 1 soat vaqt belgilash tavsiya etiladi. Natijani bilish uchun javoblaringizni javoblar varaqasiga belgilab, rasmini **@dtmxizmatbot** orqali jo‘nating. Sizga omad tilaymiz!

TEST VARIANTI RAQAMINI JAVOBLAR VARAQASIGA TO‘G‘RI KO‘CHIRING!



1. Mis va ruxdan iborat qotishmaning massasi 18 kg. Qotishmaning 60%ini rux tashkil qiladi. Qotishmaning 50%ini mis tashkil qilishi uchun unga necha kilogramm mis qo‘shish kerak?

A) *3,6* B) *3,4* C) *1,8* D) *4,2*

2. $\dfrac{3^x - p}{x-2} = 0$ tenglama yechimga ega bo‘ladigan p ning barcha qiymatlarini toping.

A) $(0;2)\cup(2;\infty)$ B) $(-\infty;9)\cup(9;\infty)$
C) $(0;9)\cup(9;\infty)$ D) $(9;\infty)$

3. $\left(x^2-4\right)^2 - 5\,(x-1)\cdot\left(x^2-4\right) - 6\,(x-1)^2 = 0$ tenglamaning barcha haqiqiy ildizlari yig‘indisini toping.

A) *4* B) *5* C) *6* D) *3*

4. Integralni hisoblang: $\int \dfrac{x^2+6x+7}{x^2+6x+10}dx$

A) $x - 3arctg\,(x+3) + C$
B) $x + 3\arcsin(x+6) + C$
C) $x - 3\arcsin(x+1) + C$
D) $x + 3arctg\,(x+3) + C$

5. Soddalashtiring: $3a - (5a - (3a - (2a + b)))$

A) $-a-b$ B) $a-b$ C) $a+b$ D) $-a+b$

6. Quyida berilgan sonlardan qaysi biri $\sqrt{3}\,\mathrm{tg}\left(\dfrac{3\pi x}{2}\right) - 1 = 0$ tenglamaning ildizi bo‘la olmaydi?

A) $2\dfrac{1}{9}$ B) $\dfrac{1}{9}$ C) $\dfrac{2}{3}$ D) $\dfrac{7}{9}$

7. $y = \sqrt{6-x} + \log_{(4-x)}\left(x^2-4\right)$ funksiyaning aniqlanish sohasini toping.

A) $(2;4)$
B) $(2;3)\cup(3;4)\cup(4;6)$
C) $(-\infty;-2)\cup(2;4)$
D) $(-\infty;-2)\cup(2;3)\cup(3;4)$

8. $A = \{x|\, x^2 \le 64,\ x \in R\}$, $B = \{x|\, x^2 > 4,\ x \in N\}$ bo‘lsa, $A \cap B$ to‘plamni aniqlang.

A) $(2;8]$ B) $[2;8]$ C) $\{2;3;4;5;6;7;8\}$
D) $\{3;4;5;6;7;8\}$

9. $f(x) = x \cdot 5^x$ funksiyaning hosilasini toping.

A) $f'(x) = 2\cdot 5^x \ln 5^x$
B) $f'(x) = 5^x \ln 5^x - 5^x$
C) $f'(x) = 5^x \ln 5^{x+1}$
D) $f'(x) = 5^x \ln 5^x + 5^x$

10. Agar $f(x) = \dfrac{-2}{x}$ bo‘lsa, u holda $y = f(x+2) + 1$ funksiyani toping.

A) $y = \dfrac{-x}{x+2}$ B) $y = \dfrac{-2x}{x+2}$ C) $y = \dfrac{2x}{2+x}$
D) $y = \dfrac{x}{x+2}$

11. $ABCD$ parallelogrammda, AD katta tomonidagi A va D burchaklarining bissektrisalari parallelogrammning BC tomonida kesishgan bo‘lsa, parallelogrammning tomonlari orasida qaysi munosabat to‘g‘ri bo‘ladi?

A) $2AB > AD$ B) $2DC > AD$
C) $2AB = AD$ D) $2AB < AD$

© Davlat test markazi, 2019

12. Beshta bir xil qog‘ozchaning har biriga quyidagi harflardan biri takrorlanmasdan yozilgan: **A, T, N, S, O**. Qog‘ozchalar qutiga solingan va yaxshilab aralashtirilgan. Qutiga qaramasdan bittalab olingan va olingan tartibda o‘qilganda **SON** so‘zi hosil bo‘lish ehtimolligini toping.

A) $\frac{1}{60}$ B) $\frac{1}{40}$ C) $\frac{1}{120}$ D) $\frac{1}{30}$

13. Tengsizlikni yeching:
$$\frac{\left(x^2+2x+1\right)(x-3)(x+4)}{x^2-4x+4}<0.$$

A) $(-\infty;-4)\cup(3;\infty)$

B) $(-\infty;-4)\cup(-1;2)\cup(3;\infty)$

C) $(-4;-1)\cup(2;3)$

D) $(-4;-1)\cup(-1;2)\cup(2;3)$

14. Asosining tomonlari 13 dm; 14 dm va 15 dm bo‘lgan uchburchakli to‘g‘ri prizmaning yon qirrasi asosining kichik balandligiga teng. Prizmaning to‘la sirtini (dm^2) toping.

A) $638,4$ B) $470,4$ C) $710\frac{10}{13}$ D) 672

15. Bitta nuqtadan tekislikka og‘ma va perpendikulyar o‘tkazilgan bo‘lib, ular orasidagi burchak 15° ga teng. Agar perpendikulyarning uzunligi $12+6\sqrt{3}$ cm bo‘lsa, og‘maning tekislikdagi proyeksiyasi uzunligini (cm) toping.

A) 12 B) 6 C) $6\cdot\left(7-4\sqrt{3}\right)$
D) $6\cdot\left(7+4\sqrt{3}\right)$

16. Quyidagi fikrlardan qaysi biri natural sonlar uchun har doim to‘g‘ri?

A) *ikkita tub sonlar yig‘indisi juft son bo‘ladi*

B) *faqat uchta natural bo‘luvchiga ega bo‘lgan son biror tub sonning kvadratiga teng bo‘ladi*

C) *barcha tub sonlar toq sonlardir*

D) *ikkita tub sonlar ko‘paytmasi toq son bo‘ladi*

17. Kasrning maxrajini irratsionallikdan qutqaring:
$$\frac{\sqrt{3}}{2\sqrt{3}-\sqrt{13}-1}$$

A) $\frac{1-2\sqrt{3}+\sqrt{13}}{4}$ B) $\frac{1-2\sqrt{3}-\sqrt{13}}{4}$
C) $\frac{1-2\sqrt{3}+\sqrt{13}}{2}$ D) $\frac{1-2\sqrt{3}-\sqrt{13}}{2}$

18. Hisoblang:
$3,6\cdot4,8+5,4\cdot3,6+4,8\cdot9,2-4,8\cdot5,6.$

A) *43,2* B) *48* C) *72* D) *54*

19. $y=kx+b$ chiziqli funksiyaning grafigi I, II choraklarda yotsa, k va b larni nol bilan taqqoslang.

A) $k=0,\ b>0$ B) $k>0,\ b=0$
C) $k<0,\ b=0$ D) $k=0,\ b<0$

20. Uchlari $A(-1;1)$, $B(3;1)$ va $C(2;4)$ nuqtalarda bo‘lgan uchburchakning yuzasini toping.

A) *12* B) *9* C) *8* D) *6*

21. $\frac{1}{10}\cdot\frac{2}{10}\cdot\frac{3}{10}\cdot\frac{4}{10}\cdot\frac{5}{10}\cdot\frac{6}{10}\cdot\frac{7}{10}\cdot\frac{8}{10}\cdot\frac{9}{10}$ ko‘paytmani standart shaklga keltiring.

A) $3,6288\cdot10^{-6}$ B) $3,6288\cdot10^{-5}$
C) $3,6288\cdot10^{-3}$ D) $3,6288\cdot10^{-4}$

22. $8\cos\frac{3x}{2}\cdot\cos\frac{x}{2}-3=6\cos2x$ tenglamaning eng kichik musbat yechimi π dan qanchaga kam?

A) $\frac{3\pi}{4}$ B) $\frac{5\pi}{6}$ C) $\frac{2\pi}{3}$ D) $\frac{3\pi}{2}$

23. To‘g‘ri burchakli uchburchakning eng kichik tomoni uzunligi 5 ga teng va qolgan tomonlari uzunliklari natural son bo‘lsa, uchburchakning yuzini toping.

A) *35* B) *6* C) *30* D) *aniqlab bo‘lmaydi*

24. $\frac{|3x-16|}{x}=-5$ tenglamaning ildizlar yig‘indisini (agar yagona bo‘lsa, ildizini) toping.

A) -8 B) *2* C) *6* D) *0*

25. $f(x)=-\frac{x^4}{5}+4x+5$ funksiya berilgan. Uning monotonligidan foydalanib, $f(\sqrt[3]{6})$ va $f(\sqrt[3]{7})$ ni taqqoslang.

A) $f(\sqrt[3]{7})-1=f(\sqrt[3]{6})+1$
B) $f(\sqrt[3]{7})=f(\sqrt[3]{6})$
C) $f(\sqrt[3]{7})<f(\sqrt[3]{6})$
D) $f(\sqrt[3]{7})>f(\sqrt[3]{6})$

26. Agar a va b haqiqiy sonlar uchun $a<0<b$ bo‘lsa, u holda $-\sqrt[3]{a^3}+\sqrt[3]{b^3}+\sqrt{a^2}-\sqrt{b^2}$ ni soddalashtiring.

A) $2b$ B) 0 C) $2b-2a$ D) $-2a$

27. Agar $b=-3$ bo‘lsa, $\frac{\left(b^{17}-1\right)(b+1)}{b^{16}+b^{15}+b^{14}+...+b+1}$ ifodaning qiymatini toping.

A) 15 B) 8 C) 4 D) 16

28. Biri ikkinchisidan 3 marta katta bo‘lgan ikki sonning yig‘indisi 9,64 ga teng. Shu sonlarning kichigini toping.

A) *2,21* B) *2,16* C) *2,31* D) *2,41*

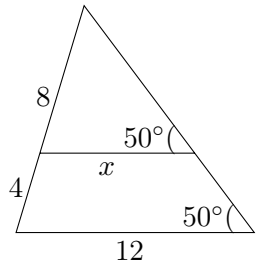
© Davlat test markazi, 2019

29. Ifodani soddalashtiring ($a \in (-2; -1)$):

$$\frac{|a^2-4|}{2-a} - \frac{|a^2-9|}{a-3} + \frac{|a^2-1|}{1-a}.$$

A) $-3\,(a+2)$ B) $-a+2$ C) $a+4$
D) $a+2$

30. Rasmdan foydalanib x ning qiymatini toping.



A) *9* B) *6,5* C) *8*
D) *berilgan ma'lumotlar yetarli emas*

Test varianti "Test topshiriqlari to'plami 2019" asosida shakllantirilgan:

1 – 21-bet 50	7 – 58-bet 39	13 – 67-bet 6	19 – 74-bet 44	25 – 84-bet 60
2 – 53-bet 9	8 – 122-bet 24	14 – 111-bet 40	20 – 116-bet 19	26 – 23-bet 10
3 – 65-bet 58	9 – 79-bet 27	15 – 108-bet 13	21 – 13-bet 34	27 – 30-bet 3
4 – 89-bet 34	10 – 70-bet 3	16 – 8-bet 16	22 – 43-bet 81	28 – 16-bet 5
5 – 45-bet 2	11 – 95-bet 38	17 – 27-bet 51	23 – 101-bet 86	29 – 49-bet 41
6 – 39-bet 50	12 – 125-bet 25	18 – 12-bet 22	24 – 61-bet 22	30 – 93-bet 28

© Davlat test markazi, 2019